EE017-6-3-PP2 Project Phase 2 – Implementation in Engineering Project

Chapter 5 : Discussion – Project Findings & Testing

Topic Learning Outcomes

At the end of this topic, you should be able to:

1. Review the sustainability and environmental measures for the implemented solution by considering social awareness, efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, with an inclusion of atleast two United Nations Sustainable Development Goals (UN-SDGs). (C5, PLO6)
2. Explain professional and ethical principles and responsibilities considered for the implemented solution by incorporating the need for diversity and inclusion. (C5, PLO7)
3. Explain project management execution, economic decision making and entrepreneurial considerations for the implemented solution. (C5, PLO10)

Topic Learning Outcomes

At the end of this topic, you should be able to:

4. Support continuous professional development by recognizing the need for independent and life-long learning with an emphasis on cutting-edge technologies through critical thinking. (C5, PLO11)

5. Defend your work done effectively and professionally through report writing and presentations by taking into account cultural, language, and learning differences such as Visual learners, Auditory learners, Kinesthetic learners, Social learners, Solo learners. (A5, PLO9)

Contents & Structure

- Testing
- Sustainable Development and Environmental
- Moral Professionalism and Ethical Consideration
- Project Management, Finance and Entrepreneurship
- Your contribution in this project

Recap From Last Lesson

- Nil

Key Terms You Must Be Able To Use

If you have mastered this topic, you should be able to use the following terms correctly in your assessments:

- Testing
- Sustainable Development
- Moral Professionalism
- Ethics
- Project Management
- Finance
- Entrepreneurship

Chapter 5 – Discussion: Project Findings & Testing

Outline

i. Testing
ii. Discrepancy between the theoretical and experimental results
iii. Explain the possible sources of error and troubleshooting methods
iv. Comparison with the previous research (Optional)
v. Sustainable Development and Environmental considerations
vi. Moral Professionalism and Ethical Consideration
vii. Project Management, Finance and Entrepreneurship 2
viii. Your contribution in this project

I. TESTING OF THE PROPOSED DESIGN

- Proposed discussion must be done with comprehensive and excellent analysis of results, problems encountered, and troubleshooting methods employed, planned testing must be carried successfully meeting all objectives without deficiencies, and incorporation of public health and safety, whole-life cost, net zero carbon, and resource, cultural, societal, and environmental considerations.

I. TESTING OF THE PROPOSED DESIGN

- List down the tests performed on the proposed design in detail.
- Example:
  - 5.1.1 Efficiency Test
    - **Definition**
    - **Experimental set up** – Explain how the system is set up in order to perform this particular test.
    - **Data Collection** – Data must be collected and tabulated. Only numerical values must be used to collect the test data, rather than saying “YES” or “NO”.
    - **Data analysis** – draw the graph or plot based on the data collected and critically give the inference based on the observation from this test.
  - 5.1.2 Test 2
    - Repeat the 5.1.1 process.

II. DISCREPANCY BETWEEN THE THEORETICAL AND EXPERIMENTAL RESULTS

- Tabulate the discrepancies between the calculated/theoretical values and simulation/hardware results and explain briefly.

III. EXPLAIN THE POSSIBLE SOURCES OF ERROR AND TROUBLESHOOTING METHODS

- Tabulate the possible sources of error and how these errors were overcome.

IV. COMPARISON WITH THE PREVIOUS RESEARCH (OPTIONAL)

- Good to compare the results obtained from this system with the system already reviewed in the literature.
- Comparison done must have the same background parameters.

V. SUSTAINABLE DEVELOPMENT AND ENVIROMENTAL CONSIDERATION

- Explain how sustainability is achieved in your project with respect to social, economic and environmental aspect. Relate the numerical values of the results achieved from the testing section to sustainability.
- Comprehensively consider social awareness, efficient resource use, environmental impacts, whole-life cost, re-use of resources, net zero carbon, and inclusion of at least eight UN-SDGs.
- Avoid explaining in general.

VI. MORAL PROFESSIONALISM AND ETHICAL CONSIDERATION

- Explain how professionalism is followed in this project duration.
- Explain the relevant codes of conduct such as BEM, FCC, ITU-T or any relevant codes related to your project and programme.
- Explain whether ethical considerations are considered or not.
- With comprehensive incorporation of the need for diversity and inclusion.
- Avoid explaining in general.

VII. PROJECT MANAGEMENT, FINANCE AND ENTREPRENEURSHIP 2

- Explain how project is managed during Phase 2 with the help of a Gantt Chart.
- Explain the actual cost incurred with the help of table.
- Explain how the project can be marketable.
- Avoid explaining in general.

VIII. YOUR CONTRIBUTION IN THIS PROJECT

- Outline briefly the contributions given in this project.
- Yes, the entire project is yours, but what is the actual contribution – such as, enhanced existing algorithm, integrating the existing algorithms, programming, new GUI design, creative ideas, and so on.

QUICK REVIEW QUESTIONS

- How many minimum testing must be done?
- Is comparison with literature is mandatory?
- What are the elements must be explained for sustainability?

SUMMARY OF MAIN TEACHING POINTS

- Testing
- Sustainable Development
- Moral Professionalism
- Ethics
- Project Management
- Finance
- Entrepreneurship

QUESTION & ANSWER SESSION

Q & A

WHAT WE WILL COVER NEXT

- Conclusion and Recommendations